Exercises for Probabilistic Inference

1. An expert system uses Bayesian reasoning to infer exhaustive hypotheses H_1, H_2 and H_3 from conditional independent evidences E_1 E_2 and E_3. The prior probabilities of the hypotheses and the conditional probabilities of observing each evidence are provided as follows:

$p(H_1) = 0.40$	$p(H_2) = 0.35$	$p(H_3) = 0.25$
$p(E_1\|H_1) = 0.3$	$p(E_1\|H_2) = 0.8$	$p(E_1\|H_3) = 0.5$
$p(E_2\|H_1) = 0.9$	$p(E_2\|H_2) = 0.0$	$p(E_2\|H_3) = 0.7$
$p(E_3\|H_1) = 0.6$	$p(E_3\|H_2) = 0.7$	$p(E_3\|H_3) = 0.9$

   (1) Evaluate the posterior probabilities p(H1|E1), p(H2|E1) and p(H3|E1) after evidence E1 is observed.
   (2) Evaluate the posterior probabilities p(H1|E1E2), p(H2|E1E2)and p(H3|E1E2) after both evidences E1 and E2 are observed.
   (3) Evaluate the posterior probabilities p(H1|E1E2E3), p(H2|E1E2E3) and p(H3|E1E2E3) after all three evidences are observed.

2. In a fault diagnosis expert system, one needs to infer exhaustive hypotheses H1 (faulty) and H2 (not faulty) from observation E (starter noisy). From empirical studies, one obtains the following conditional probabilities:
   p(E|H1) = 0.85
   p(E|H2) = 0.20
   An expert is also consulted, who provides the following judgement:
   p(H1) = 0.05
   p(H1|E) = 0.70

   (1) Evaluate the posterior probabilities p(H1|E) from the empirical information (p(E|H1) and p(E|H2)) and the prior provided by the expert (p(H1));
   (2) Verify whether the expert's judgement is consistent from your result, and explain its implications in knowledge acquisition when developing an expert system.

3. A fault diagnosis system uses Bayesian reasoning to infer exhaustive hypotheses H1 (faulty) and H2 (not faulty) from evidences E1 (noisy) and E2 (overheating). The prior probabilities of the hypotheses and the conditional probabilities of observing each evidence are provided as follows.

p(H1) = 0.1	p(H2) = 0.9
p(E1|H1) = 0.8	p(E1|H2) = 0.3
p(E2|H1) = 0.9	p(E2|H2) = 0.1

   If we assume evidences E1 and E2 are independent:
   (1) Evaluate the posterior probabilities p(H1|E1) and p(H2|E1) after evidence E1 is observed;
   (2) Evaluate the posterior probabilities p(H1|E1E2) and p(H2|E1E2) after both evidences E1 and E2 are observed.

4. The Monty Hall problem is based on an American TV game show. It can be described as follows. Behind three closed doors, there is a car (the prize) and two goats (worthless). You pick a door, say No. 1. Then the host (Monty Hall in the show), who knows what are behind the doors, opens another door that has a goat, say No. 3. Now you have the chance to decide to stick to your original choice of No. 1, or to switch to No. 2.

   Using a Bayesian Network, evaluate the chance of winning for sticking to original choice 1 and switching to 2.